Rethinking with Retrieval: Faithful Large Language Model Inference

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Problem

LLMs are good **but**

Incomplete – out of date informations

Chain Of Thought Issues

Cost

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Offered Solution

Rethinking With Retrival (RR)

Given a query Q ,

Utilize CoT prompting to generate a diverse set of reasoning paths $R_1, R_2, \cdots R_N$

Each reasoning path R_i consists of an explanation E_i followed by a prediction P_i

After that, we retrieve relevant knowledge $K_1, \cdots K_m$ from a suitable knowledge base KB to support the explanation in each reasoning path

Select the Prediction that is most faithful to this knowledge

Q: Did Aristotle use a laptop?

CoT Prompting: It aims to generate the complete reasoning path.

"Aristotle died in 322 BC.
The first laptop was invented in 1980.
Thus, Aristotle did not use a laptop.
So the answer is no."

Empirical results show that CoT prompting significantly improves the performance of LLMs on many multistep reasoning tasks. Therefore, we adopt CoT prompting to obtain both explanation E and prediction P for the query Q.

Sampling diverse reasoning paths: We sample a diverse set of reasoning paths $R_1, R_2, \cdots R_N$ rather than only considering the greedy path

(R1) Aristotle died in 2000. The first laptop was invented in 1980. Thus, Aristotle used a laptop. So the answer is yes.

(R2) Aristotle died in 322BC. The first laptop was invented in 2000. Thus, Aristotle did not use a laptop. So the answer is no.

(R3) Aristotle died in 322BC. The first laptop was invented in 1980. Thus, Aristotle did not use a laptop. So the answer is no.

Knowledge retrieval: Different knowledge bases can be used to address different tasks. For example, to address the question “Did Aristotle use a laptop?”, we can use Wikipedia as the external knowledge base.

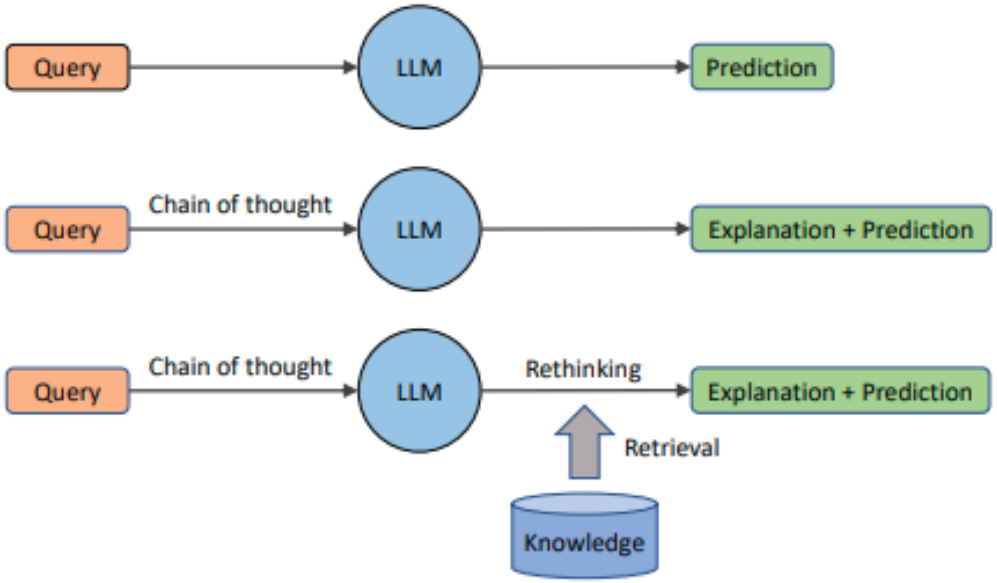
Ideally, we would obtain the following two paragraphs from Wikipedia for this question:

(K1) Aristotle (384–322 BC) was a Greek philosopher and polymath during the Classical period in Ancient Greece. ...

(K2) The Epson HX-20, the first laptop computer, was invented in 1980. ...

Faithful inference: The faithfulness of each reasoning path R_i can be estimated using a function $f_{KB}(R_i)$, which is based on relevant knowledge $K_1, \cdots, K_M$ retrieved from the knowledge base.

$$\hat{P} = \underset{P_i \in \{P_1, \cdots, P_N\}}{\arg\max} \sum_{i=1}^{N} \mathbb{1}(P_i = P) f_{\mathcal{KB}}(R_i),$$


Query
LLM
Prediction
Query
Chain of thought
LLM
Explanation + Prediction
Query
Chain of thought
LLM
Rethinking
Explanation + Prediction
Retrieval
Knowledge

Benefits

- No training
- No fine-tuning
- Not limited by context window

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Experiment Setup

Baselines

Zero-shot/few-shot prompting: In our experiments, we consider GPT-3 with standard zeroshot/few-shot prompting

Chain-of-thought prompting: In addition to the standard zero-shot/few-shot prompting
we also consider GPT-3 with the CoT prompting as a baseline in experiments.

Self-consistency: We also consider self-consistency
as a baseline in our experiments.

Commonsense Reasioning

Dataset description:

StrategyQA.

Includes questions that require implicit reasoning strategies.

"Did Aristotle use a laptop?" requires implicit decomposition into reasoning steps

"Was Aristotle alive when the laptop was invented?" explicitly specifies the reasoning process.

Implementation details:

Wikipedia as the external knowledge base

Retrieval: BM25 (top 10 paragraphs)

Reranking: MPNet similarity

NLI model: **entailment / contradiction scores**

Temporal Reasioning

Dataset description:

TempQuestions dataset

Includes Temporal questions that are divided into four classes: explicit temporal, implicit temporal, temporal answer, and ordinal constraints

The questions are paired with their answers from Freebase

Who was governor of oregon when shanghai noon was released?

Implementation details:

Wikidata as the external knowledge base

Entity linking: spaCy entity linker

Extract: “temporal relations” from entity pages
Convert relations to sentences with templates

Tabular Reasioning

<u>Dataset description:</u>

INFOTABS dataset

Includes 23, 738 human written textual hypotheses based on premises in the form of tables extracted from 2, 540 unique Wikipedia info-boxes

<u>Implementation details:</u>

WordNet + ConceptNet as external knowledge base

Tables converted to text

Retrieve word relation triples

Convert triples into sentences

Combine triples + table text

For the evaluation:

The model is GPT-3 text-davinci-002

The maximum number of tokens for generation is 256

For zero-shot, few-shot, and chain-of-thought prompting, the temperature is fixed at 0.

For self consistency and rethinking with retrieval, researchers randomly sample 10 outputs with temperature 0.7

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Results

	Methods	Commonsense	Temporal	Tabular
GPT-3	Zero-shot prompting	58.08	28.40	82.00
	Few-shot prompting	63.32	29.59	83.08
	Chain-of-thought prompting	65.94	33.14	83.33
	Self-consistency	73.36	37.28	84.00
	Rethinking with retrieval	**77.73**	**39.05**	**84.83**

Table 1: Performance of different methods using GPT-3 on three reasoning tasks.

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Analysis

Limitations of LLMs in Reasoning

- Wrong Supporting Facts
- Wrong Inference

	Wrong supporting facts	Wrong inference
Question	Was Lil Jon's top ranked Billboard song a collaboration with a member of The Lox?	Would the top of Mount Fuji stick out of the Sea of Japan?
Answer	No	Yes
Facts	Lil Jon's highest ranked billboard song was Yeah; Yeah was a collaboration between Lil Jon, Usher, and Ludacris; The Lox is a rap trio consisting of: Styles P, Sheek Louch, and Jadakiss.	The average depth of the Sea of Japan is 5,748 feet (1,752 metres) and its maximum depth is 12,276 feet (3,742 metres); Mount Fuji is 3,776.24 metres (12,389.2 ft) tall.
GPT-3	Lil Jon's top ranked Billboard song was "Get Low" with the East Side Boyz. The Lox is not a member of the East Side Boyz. Thus, Lil Jon's top ranked Billboard song was not a collaboration with a member of The Lox. So the answer is no.	Mount Fuji is 3,776 meters tall. The Sea of Japan is about 3,741 meters deep. Thus, the top of Mount Fuji would not stick out of the Sea of Japan. So the answer is no.

Table 2: Examples of incorrect outputs from GPT-3 with CoT prompting.

Ablation Studies

Retrieval	Commonsense	Tabular
Query-based	73.36	36.69
Decomposition-based	**77.73**	**39.05**

Table 3: Comparison of query-based and decomposition-based retrieval on commonsense and tabular reasoning.

Using the decomposed CoT steps instead of the original query improves retrieval performance.

Motivation for Variants

The basic RR approach weights entire reasoning paths and applies a voting mechanism to select the most faithful prediction.

But, CoT outputs often contain:

- irrelevant or noisy statements
- missing supporting facts
- steps that are only weakly aligned with external knowledge.

Fact Selection and **Fact Generation**

Fact Selection

Goal: Improve explanation consistency by selecting the most reliable facts from all sampled reasoning paths

- Sentences across all CoT outputs are clustered using MPNet sentence embeddings.
- Each cluster represents a specific “topic group.”
- The sentence with the highest faithfulness score within each cluster is selected as the canonical fact.

Fact Generation

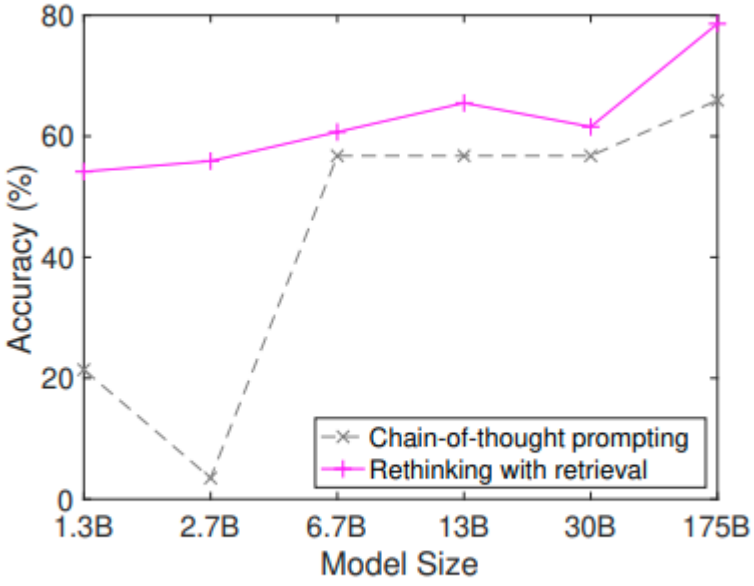
Goal: Create corrected or missing facts using external knowledge when CoT outputs are incomplete or inaccurate.

- Generate targeted questions for named entities using a Question Generation model.
- Use UnifiedQA to retrieve correct answers from the knowledge base.
- Convert the QA pairs into declarative factual statements.
- Replace unsupported or incorrect CoT sentences with these newly generated facts.

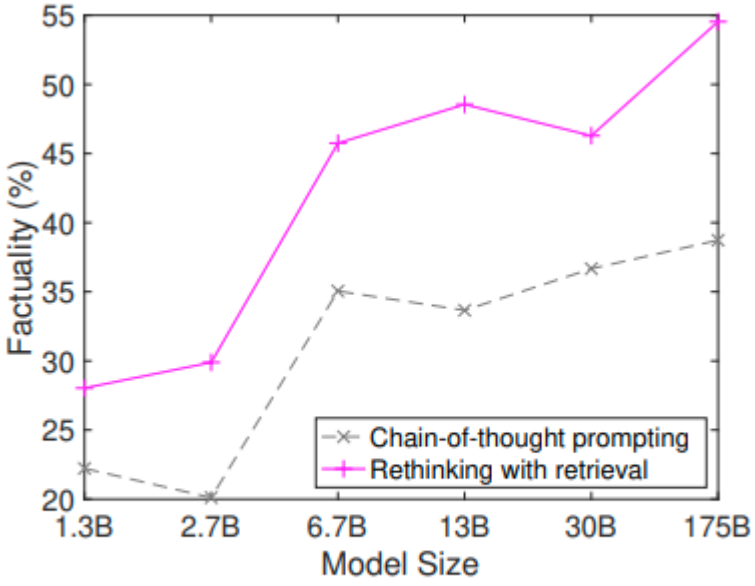
Methods	Accuracy (%)	Faithfulness (%)
CoT prompting	65.94	38.73
Basic (w/o voting)	76.86	50.02
Variant I	**78.60**	54.11
Variant II	**78.60**	**54.54**

Table 5: Comparison of various variations of RR and the CoT prompting baseline on StrategyQA using evidence paragraphs.

Impact of Model Size



(a) Accuracy of predictions



(b) Faithfulness of explanations

Conclusion

- RR provides a lightweight and effective way to integrate external knowledge into LLM inference.
- Experiments across three reasoning tasks (commonsense, temporal, tabular) show that RR consistently improves both accuracy and the faithfulness of explanations.
- RR demonstrates that external knowledge can systematically correct and refine CoT reasoning

Thanks!

Do you have any questions?